

Campus Food Delivery Application

Domain: Food

1. General Problem Description

This project focuses on food delivery services within a college campus. Students and faculty often have limited time between classes, meetings, and labs, making it difficult to access nearby dining options.

The business provides a campus-focused food delivery service that allows customers to place food and drink orders from approved campus and nearby restaurants and schedule deliveries to specific campus buildings. Orders are delivered by student or faculty, as “runners”, enabling fast delivery while creating on-campus employment.

The system operates over an academic year, and business data is analyzed at the end of each quarter to evaluate performance, customer demand, restaurant usage, and delivery efficiency.

2. Data to Be Captured

This section defines the domain and scope of the business problem being addressed. It explains the need for a campus-specific food delivery service, the environment in which it operates, and the overall purpose of collecting business data for quarterly analysis and decision-making.

Major Participants and Their Data

This subsection lists the key entities involved in the business, such as customers, runners, restaurants, and campus locations. For each participant, it describes the attributes needed to understand their role and behavior within the business. IDs are used to uniquely link customers, runners, restaurants, menu items, and locations across orders and deliveries, ensuring accurate reporting.

1) Users

- User ID
- Phone#
- Email
- Address
- Name

2) Customers (Students and Faculty)

- User ID
- Customer ID
- Age Group
- Role (Student / Faculty)

3) Runners (Delivery Employees)

- User ID
- Runner ID
- Role (Student / Faculty)
- Age Group

4) Restaurants / Vendors

- Restaurant ID
- Restaurant name
- Vendor type (On-campus / Off-campus)
- Distance from campus center
- Menu items

5) Menu Items

- Item ID
- Item name
- Item category (Food / Drink)
- Item price

6) Student

- Age Group

7) Faculty

- Age Group

8) Shifts

- Shift ID
- Actual hours
- Scheduled hours
- Date, day, and time of shift

9) Orders

- Order ID
- Payment method
- Delivery status
- Order status
- Actual delivery time
- Scheduled delivery time
- Date
- Cost
- Tip

10) Campus Locations

- Campus location ID
- Building name
- Building type (academic, lab, dorm, etc.)
- Building address

The system models a transactional service where customers place orders from restaurants, orders contain menu items, runners deliver orders during shifts, and deliveries occur at campus locations.

Interactions and Data Captured

This subsection describes how participants interact with one another, such as customers placing orders and runners delivering them. For each interaction, it specifies the data captured as part of that interaction, ensuring the system can support meaningful reporting and analysis. Customers place food and drink orders with restaurants for delivery to specific campus locations. Runners are assigned to deliver these orders during their scheduled work hours, capturing delivery time, distance, and tips. These interactions connect customers, restaurants, runners, and campus buildings to support quarterly business analysis.

1) Customers Place Orders

Captured data:

- Order ID
- Customer ID
- Restaurant ID
- Order date and time
- Scheduled delivery time
- Delivery location (building)
- Total order cost
- Order status (placed, assigned, delivered)
- Payment method (card, campus dining credit, etc.)

2) Orders Contain Menu Items

Captured data:

- Order ID
- Item ID
- Quantity
- Item price at time of order

3) Runners Deliver Orders

Captured data:

- Order ID
- Runner ID
- Scheduled delivery time
- Actual delivery time
- Delivery status (on time, delayed, incomplete)
- Delivery duration (derived)
- Distance traveled (miles)
- Tip amount

4) Restaurants located at Campus Locations

Captured data:

- Campus ID
- Building type
- Building address

- Building name
- Restaurant name
- Restaurant ID
- Distance from Campus center

5) Restaurants offer Menu Items

Captured data:

- Restaurant ID
- Item ID
- Restaurant name
- Menus
- Prices

6) Orders Deliver to Campus Locations

Captured data:

- Order ID
- Campus Location ID
- Building Name
- Building Address

7) Runners work Shifts

Captured data:

- Runner ID
 - Work date
 - Schedule work hours
 - Actual work hours
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3. Assumptions

This section documents the explicit assumptions made about how the business operates. These assumptions clarify the boundaries of the system and prevent ambiguity during database design and future analysis.

- 1) Each order is assigned to at most one runner and may remain unassigned if no runner is available.
 - 2) Menu prices may change, but the price at order time is recorded.
 - 3) Deliveries only occur within defined campus buildings.
 - 4) A quarter is the primary unit of business analysis.
 - 5) Distance is analyzed based on restaurants distance from the campus center
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4. Business Goals / Functional Requirements

This section defines the reports and aggregate information the business owner wants to generate, typically at the end of each quarter. These goals focus on insightful, non-operational analysis that helps evaluate performance, identify trends, and guide decisions for future quarters.

- 1) Identify restaurants whose total number of orders is above the average number of orders per restaurant.
- 2) Determine whether restaurants farther from campus center receive fewer orders than closer restaurants.
- 3) Identify the age group that placed the highest number of orders.
- 4) Identify the one-hour time interval with the maximum number of orders.
- 5) Determine the number of orders that could not be assigned to runners, grouped by time of day.
- 6) Identify the top 5 runners by:
 - Number of completed deliveries
 - Total tips earned
- 7) Identify the top 5 most frequently ordered food items and top 5 drink items during the quarter.

- 8) Identify the campus buildings with the highest number of deliveries.
 - 9) Determine the number and percentage of repeat customers, define as customers who placed more than one order in the same quarter.
 - 10) Compare quarter-to-quarter performance, including:
 - Total revenue
 - Average order value
 - Total number of orders
 - Runner utilization rate (assigned delivery time ÷ scheduled shift time)
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Team Information:

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